TRANSLATIONS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: MORLEYIZATION AS A SAMENESS OF THEORIES, ITS JOIN WITH MORITA EQUIVALENCE, AND THE FORCED ORDER OF TRANSLATION AND COMPLETION

RYOTA SHIBUSAWA

Abstract. The Morleyization, which presents a classical first-order theory as a coherent one, is the reflector B of the 2-category of coherent categories onto the Boolean coherent categories. We read it, as the completions F, E, P were read before, as the class V_{B} of interpretations it inverts in the lattice of two-out-of-three classes: two coherent theories are V_{B}-equivalent when their Morleyizations have equivalent syntactic categories. The join $V_{\mathsf{B}} \vee V_{\mathsf{P}}$ of the classical and the Morita samenesses is the class inverted by the Boolean pretopos completion PB, the classical imaginaries with finite coproducts, and is reached by one alternation, translation first. By the Makkai–Reyes conceptual completeness theorem this join is exactly the class of interpretations inducing an equivalence of the categories of models with elementary embeddings, whereas homomorphisms detect the Morita class alone. The order is forced: completing first and Morleyizing second fails for the coherent theory of complete multipartite graphs, where the set of parts, a classical imaginary, is not a classical subobject of any coherent imaginary uniformly in the models. The two orders agree iff, in the saturated models of the completions, every classical imaginary has the automorphism stabiliser of a coherent imaginary, Poizat's criterion relative to the coherent imaginaries; weak elimination of imaginaries for the Morleyization suffices, and a criterion in terms of coherent types gives the failures. For theories without axioms over a relational signature the orders agree iff the signature is monadic.

1. Introduction

Translations between logics present a theory of one logic as a theory of another: the double negation translations of classical into intuitionistic logic, the Gödel–McKinsey–Tarski translation of intuitionistic logic into $\mathsf{S4}$, and

2020 *Mathematics Subject Classification.* 03G30 (primary), 18F10, 03C07, 03C40 (secondary).

Key words and phrases. Morleyization, Morita equivalence, pretopos completion, elimination of imaginaries, conceptual completeness, elementary embedding, two-out-of-three, coherent category, translation of logics.

Corresponding author. E-mail: r-shibusawa@daiichi-koudai.ac.jp.

the Morleyization, which presents a classical first-order theory T as a coherent theory T^m by naming every first-order formula with a relation symbol [9, D1.5.13]. Categorically the Morleyization is a reflector: the 2-category **Coh** of small coherent categories and coherent functors has the full sub-2-category **BCoh** of Boolean coherent categories, those in which every subobject has a complement, and $T \mapsto T^m$ is the left adjoint $\mathsf{B}\colon \mathbf{Coh} \to \mathbf{BCoh}$ of the inclusion (Section 3). In [18] the completions F (finite coproducts), E (quotients of equivalence relations) and $\mathsf{P} = \mathsf{EF}$ (pretopos completion) were read as points of the lattice Same(**Coh**) of two-out-of-three classes of interpretations, each completion R defining the class V_R of interpretations it inverts, and Morita equivalence [1, 19] was identified with V_P, the join $V_\mathsf{F} \vee V_\mathsf{E}$ reached by one alternation with coproducts first. This paper reads the translation B in the same lattice.

The class V_B is a *classical sameness*: two coherent theories are V_B-equivalent when their Morleyizations are equivalent, that is, when they have the same classical first-order content, whatever their coherent content. It is incomparable with the Morita sameness V_P (Example 3.5). The results are the following.

(A) $V_\mathsf{B} \vee V_\mathsf{P} = V_\mathsf{PB}$, the class inverted by the Boolean pretopos completion, and the join is reached by one alternation, translation first: two coherent theories with Morita equivalent Morleyizations are connected by one Morleyization, one Morita equivalence and back (Theorem 4.1). The Boolean pretopos completion of $\mathcal{C}_T$ adds to the Morleyization of T its imaginaries and finite coproducts of sorts; it is Shelah's T^{eq} under the hypotheses of [6], that every sort is provably inhabited and some sort has two provably distinct elements.

(B) V_PB is the class of interpretations inducing an equivalence of the categories of models with *elementary embeddings*, by the Makkai–Reyes conceptual completeness theorem [11, 7.1.8] applied to PB (Theorem 4.3). Since the class inducing an equivalence of the categories of models with *homomorphisms* is V_P [18, Thm. 6.1], the choice of morphisms between models decides whether the classical sameness is seen: homomorphisms detect Morita equivalence, elementary embeddings detect Morita equivalence up to Morleyization.

(C) The order of the alternation is forced. Completing first and Morleyizing second gives $\mathsf{BP}\mathcal{C}$, a full subcategory of $\mathsf{PB}\mathcal{C}$, which is all of it iff it is a pretopos (Lemma 5.1); this holds for Boolean and for propositional theories (Proposition 5.2) but not in general: in the coherent theory of complete multipartite graphs the quotient of the sort by the complement of the edge relation, the set of parts, is a classical imaginary which is not a classical subobject of any coherent imaginary (Theorem 5.3). The proof counts fixed points of automorphisms of the complete k-partite graphs, as the sharpness proof of [18, Thm. 5.6] counted points at finitely presented models;

the obstruction is one of uniformity, since for each k the parts of the k-partite models are a coherent quotient, with $k-1$ witnesses.

(D) The commutation $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ is characterised (Theorem 6.1): it holds iff in the saturated models of the completions of T^m every classical imaginary has the automorphism stabiliser of a coherent imaginary of T, which is Poizat's Galois criterion for elimination of imaginaries [12] relativised to the coherent imaginaries. Hence weak elimination of imaginaries for T^m implies commutation (Corollary 6.4), and a criterion in terms of coherent types (Proposition 6.2) gives the failures without saturation; it reproves (C) in one infinite model.

(E) For the coherent theories without axioms over a relational signature the two orders agree iff the signature is monadic (Theorem 7.1).

(F) The orders agree for the coherent theory of an equivalence relation (Theorem 8.2), which has the same Morleyization as the theory of complete multipartite graphs: the commutation is a property of the coherent presentation and not of the classical theory (Corollary 8.3).

(G) The Morleyization is the Heyting completion followed by the addition of excluded middle, and the second commutes with the pretopos completion for every Heyting category (Theorem 9.2): the forced order arises in the Heyting completion, the passage to implication and the universal quantifier, not in excluded middle (Corollary 9.3).

What is known: the Morleyization and its universal property [9, D1.5.13], also for invertible 2-cells [5, §6.2, Prop. 6.5]; that a classical theory and its Morleyization have the same Morita class and that the models of T^m are the models of T with elementary embeddings [19, §5], [7]; the pretopos completion as T^{eq} under Harnik's hypotheses [6]; weak elimination of imaginaries for theories with unary relations only [2, Rem. 7.14]; conceptual completeness [11]; the Galois criterion for elimination of imaginaries and its weak form [12, 4]; the lattice and the join lemma [16, 18]. So (B) is a direct specialisation of [11] and the equivalence (i)⇔(iii) of (D) relativises a known criterion. What is new is the reading of a translation as a sameness and the join (A) in the lattice, the non-commutation (C) with the multipartite example, the relative form of the criterion with the coherent-type test of (D), and the dichotomy (E), of which the failure for every symbol of arity at least two is the new half, (F) and (G). The last section lists the next translations to be read in the same way.

2. The lattice, the completions and the join lemma

We recall from [16, 18] what is used. A *sameness* on a 2-category is a class W of 1-cells containing the equivalences, closed under composition and satisfying two-out-of-three up to invertible 2-cells; $\mathrm{Same}(\mathcal{C})$ is the complete lattice of samenesses under inclusion. For a 2-functor R, V_R denotes the

class of 1-cells f with Rf an equivalence, the largest sameness R inverts. Two objects are *W-equivalent* if connected by a zig-zag of 1-cells in W.

Coh is the 2-category of small coherent categories (finite limits, images, finite unions of subobjects, all stable under pullback), coherent functors [9, A1.4] and *natural isomorphisms* as 2-cells. Only invertible 2-cells enter the definition of Same(**Coh**), so nothing is lost by this restriction, and it is needed: the Morleyization is not functorial on non-invertible natural transformations (Remark 3.3). A coherent theory T has a syntactic category $\mathcal{C}_T$, interpretations $T \to T'$ are coherent functors $\mathcal{C}_T \to \mathcal{C}_{T'}$, and every coherent category is the syntactic category of its internal theory [9, D1.4], so we identify theories with coherent categories. A coherent category is *positive* if it has finite coproducts, disjoint and stable, *effective* if every equivalence relation is a kernel pair, a *pretopos* if both; **Pretop** $\subseteq$ **Coh** is reflective with reflector P, the pretopos completion [9, A3.3], $\mathsf{P} = \mathsf{EF}$ with F the positive and E the exact completion of a regular category (ex/reg) [3, 10]. Set-models of $\mathcal{C}$ are coherent functors $\mathcal{C} \to$ **Set** and $\mathrm{Mod}(\mathcal{C})$ is their category with natural transformations, the homomorphisms of models; $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$ by the universal property. Morita equivalence of coherent theories in the sense of [1, 19] is V_{P}-equivalence [19, Thm. 4.7, Cor. 4.9], and $V_{\mathsf{P}} = V_{\mathsf{F}} \vee V_{\mathsf{E}}$ by the following lemma [18, Lemma 2.3, Thm. 4.1].

Lemma 2.1 (join lemma)**.** *Let R_1, R_2 be reflectors of a 2-category onto reflective sub-2-categories, with units η_i, and suppose that $R := R_2R_1$ is again a reflector, with unit $X \to R_1X \to R_2R_1X$, and that R inverts the units of R_1 and of R_2. Then $V_R = V_{R_1} \vee V_{R_2}$, and for every $f\colon X \to Y$ in V_R the objects X and Y are connected by the zig-zag of one R_1-unit, one R_2-unit, the equivalence Rf, and one R_2-unit and one R_1-unit backwards.*

3. Morleyization as a reflector

Definition 3.1. A coherent category is *Boolean* if every subobject has a complement. **BCoh** $\subseteq$ **Coh** is the full sub-2-category of Boolean coherent categories and **BPretop** $=$ **BCoh** $\cap$ **Pretop** that of Boolean pretoposes.

A Boolean coherent category is a Heyting category, $\forall_f S = \neg\exists_f\neg S$, so every first-order formula over its internal theory is interpreted in it, and coherent functors between Boolean coherent categories preserve complements, hence all first-order formulas; thus **BCoh** is full in **Coh** and the internal theories of Boolean coherent categories are the classical first-order theories.

Proposition 3.2. *The inclusion* **BCoh** $\to$ **Coh** *has a left biadjoint* B*, with* $\mathbf{Coh}(\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathsf{B}\mathcal{C}, \mathcal{B})$ *as groupoids for $\mathcal{B}$ Boolean coherent, and $\mathsf{B}\mathcal{C}_T = \mathcal{C}_{T^m}$ the syntactic category of the Morleyization of T: the theory in the signature of T enlarged by a relation symbol R_φ for every first-order formula φ over that signature, with the coherent axioms making R_φ equivalent to φ (the axioms of* [9, D1.5.13]*). The unit $\eta\colon \mathcal{C} \to \mathsf{B}\mathcal{C}$ is the inclusion of the coherent formulas.*

Proof. $\mathcal{C}_{T^m}$ is Boolean: $R_{\neg\varphi}$ is a complement of R_φ and every object is a subobject of a product of sorts cut out by some R_φ. For $\mathcal{B}$ Boolean coherent, a coherent functor $\mathcal{C}_T \to \mathcal{B}$ is a model of T in $\mathcal{B}$; it interprets every first-order formula φ in $\mathcal{B}$, and setting R_φ to that interpretation satisfies the axioms of T^m, giving a model of T^m, uniquely since the axioms determine R_φ; an isomorphism of models of T in $\mathcal{B}$ preserves the interpretations of all first-order formulas, so it is an isomorphism of the models of T^m. So $\mathbf{Coh}(\mathcal{C}_T, \mathcal{B}) \simeq \mathbf{Coh}(\mathcal{C}_{T^m}, \mathcal{B})$ by restriction along η. □

Remark 3.3. With all natural transformations as 2-cells the statement fails. Let $\mathcal{C} = \{\bot < p < \top\}$ and $\mathcal{D} = \{0, p, \neg p, 1\}$ the four-element Boolean algebra, and let $F, G\colon \mathcal{C} \to \mathcal{D}$ be the coherent functors with $F(p) = 0$, $G(p) = 1$. Since $F \leq G$ pointwise there is a natural transformation $F \Rightarrow G$, but $\mathsf{B}F(\neg p) = 1 \not\leq 0 = \mathsf{B}G(\neg p)$, so it does not extend to $\mathsf{B}\mathcal{C} \simeq \mathcal{D}$. A homomorphism of models of T need not be a homomorphism of models of T^m: the latter are the elementary embeddings (Lemma 4.2). The pretopos completion, by contrast, is a 2-reflection for all natural transformations [9, A3.3], which is what Lemma 4.2 uses.

Conventions. For a coherent theory T, $\mathcal{C}_T$ is its coherent syntactic category and $\mathsf{B}\mathcal{C}_T$ the first-order syntactic category of the same axioms read classically; a classical theory is identified with its Morleyization, so that for it $\mathcal{C}_T$ is Boolean and B is the identity. Formulas are without parameters; sorts may be empty. An equivalence $\mathsf{BP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$ always means that the canonical comparison functor of Lemma 5.1 is an equivalence.

We write V_B for the class of coherent functors inverted by B and call V_B-equivalence the *classical sameness*: $\mathcal{C}$ and $\mathcal{D}$ are classically the same when their Morleyizations have equivalent syntactic categories, whatever the coherent fragments.

Proposition 3.4. P *preserves Booleanness,* PB *is the reflector of* **Coh** *onto* **BPretop**, *and* $\mathsf{PBP} \simeq \mathsf{PB}$. *The objects of* $\mathsf{PB}\mathcal{C}_T$ *are the quotients of finite coproducts of classical definable sets by classical equivalence relations; under the hypotheses of* [6]*, that every sort is provably inhabited and some sort has two provably distinct elements (*$T^m \vdash \exists x{:}A\, \top$ *for every sort* A *and* $T^m \vdash \exists u, v{:}A_0\, (u \neq v)$ *for some sort* A_0*; no constants are required), this is the syntactic category of Shelah's* $(T^m)^{\mathrm{eq}}$*, and in general it is larger.*

Proof. In $\mathsf{F}\mathcal{B}$ a subobject of $X_1 + \cdots + X_n$ is a sum of subobjects, complemented by the sum of the complements; in $\mathsf{EF}\mathcal{B}$ a subobject of X/E is S/E for an E-saturated $S \subseteq X$, and the complement of a saturated subobject is saturated. So $\mathsf{P}\mathcal{B}$ is Boolean for $\mathcal{B}$ Boolean. Hence $\mathsf{PB}\mathcal{C} \in$ **BPretop**, and for $\mathcal{B} \in$ **BPretop**, $\mathbf{Coh}(\mathsf{PB}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathsf{B}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{B})$: PB is the reflector. Both $\mathsf{PBP}\mathcal{C}$ and $\mathsf{PB}\mathcal{C}$ represent $\mathbf{Coh}(\mathcal{C}, -)$ on **BPretop**, since $\mathbf{Coh}(\mathsf{PBP}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathsf{P}\mathcal{C}, \mathcal{B}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{B})$, so the canonical functor $\mathsf{PB}\mathcal{C} \to \mathsf{PBP}\mathcal{C}$ is an equivalence. The description of the objects is that of $\mathsf{P} = \mathsf{EF}$, and the identification with T^{eq} under the two hypotheses is [6];

without them it can fail, coproduct sorts not being definable: for the classical theory of a one-element sort, T^{eq} has only sorts with at most one element, while $\mathsf{PB}\mathcal{C}_T \simeq \mathbf{FinSet}$ contains $1+1$. □

(1)
$$\begin{array}{ccc} \mathbf{Coh} & \xrightarrow{\ \mathsf{B}\ } & \mathbf{BCoh} \\ {\scriptstyle \mathsf{P}}\downarrow & & \downarrow{\scriptstyle \mathsf{P}} \\ \mathbf{Pretop} & \overset{\mathsf{B}}{\dashrightarrow} & \mathbf{BPretop} \end{array}$$

The composite along the top and right is the reflector PB; the diagram is not a commutative square, since the dashed arrow does not land in **BPretop** in general (Theorem 5.3): B of a pretopos need not be a pretopos.

Example 3.5 (incomparability)**.** (i) $V_{\mathsf{B}} \not\subseteq V_{\mathsf{P}}$. Let $\mathcal{C}$ be the three-element chain $\bot < p < \top$, the syntactic category of the coherent propositional theory with one atom, and $\mathcal{D} = \{0, p, \neg p, 1\}$ the four-element Boolean algebra. The inclusion $\mathcal{C} \to \mathcal{D}$ is inverted by B, both sides Morleyizing to $\mathcal{D}$, but not by P: $\mathsf{P}\mathcal{C}$ is the category of coherent objects of the Sierpiński topos and $\mathsf{P}\mathcal{D} \simeq \mathbf{FinSet} \times \mathbf{FinSet}$, and they are not equivalent (the former has a non-complemented subterminal). (ii) $V_{\mathsf{P}} \not\subseteq V_{\mathsf{B}}$. Let $\mathbf{2} = \{\bot < \top\}$ be the initial coherent category, the empty theory on no sorts, and $\mathbf{2} \to \mathsf{F}\mathbf{2} = \mathbf{FinSet}$ the unit of F, inverted by P. $\mathsf{B}\mathbf{2} = \mathbf{2}$ and $\mathsf{B}\mathbf{FinSet} = \mathbf{FinSet}$ are not equivalent.

4. The join and the models with elementary embeddings

Theorem 4.1 (the join, translation first)**.** *In* Same(**Coh**)*, $V_{\mathsf{PB}} = V_{\mathsf{B}} \vee V_{\mathsf{P}}$: the class of interpretations inverted by the Boolean pretopos completion is the join of the classical and the Morita samenesses, and for every $f\colon \mathcal{C} \to \mathcal{D}$ in it the two theories are connected by the zig-zag*

$$\mathcal{C} \to \mathsf{B}\mathcal{C} \to \mathsf{PB}\mathcal{C} \xrightarrow{\simeq} \mathsf{PB}\mathcal{D} \leftarrow \mathsf{B}\mathcal{D} \leftarrow \mathcal{D}.$$

Hence two coherent theories have Morita equivalent Morleyizations iff they are connected by one classical sameness and one Morita equivalence on each side, in this order.

Proof. Lemma 2.1 with $R_1 = \mathsf{B}$, $R_2 = \mathsf{P}$, $R = \mathsf{PB}$, which is a reflector by Proposition 3.4. R inverts the units of B, since $\mathsf{BB} = \mathsf{B}$, and those of P, since $\mathsf{PB}\mathcal{C} \to \mathsf{PBP}\mathcal{C}$ is an equivalence (Proposition 3.4). □

Let $\mathrm{Mod}_{\mathrm{el}}(\mathcal{C})$ be the category of Set-models of $\mathcal{C}$ with the *elementary embeddings*, the natural transformations whose components preserve and reflect every first-order formula over the internal theory of $\mathcal{C}$.

Lemma 4.2. $\mathrm{Mod}_{\mathrm{el}}(\mathcal{C}) \simeq \mathrm{Mod}(\mathsf{B}\mathcal{C}) \simeq \mathrm{Mod}(\mathsf{PB}\mathcal{C})$.

Proof. Models of $\mathsf{B}\mathcal{C}$ are models of $\mathcal{C}$ (Proposition 3.2 with $\mathcal{B} = \mathbf{Set}$). A homomorphism of models of $\mathsf{B}\mathcal{C}$ preserves every subobject of $\mathsf{B}\mathcal{C}$, hence every first-order formula and its negation, so it is an elementary embedding; conversely an elementary embedding is natural on every R_φ. The second equivalence is the universal property of P, which holds for all natural transformations [9, A3.3]. □

Theorem 4.3 (conceptual completeness for elementary embeddings)**.** *$V_{\mathrm{Mod_{el}}} = V_{\mathsf{PB}}$: an interpretation $f\colon \mathcal{C} \to \mathcal{D}$ induces an equivalence $\mathrm{Mod_{el}}(\mathcal{D}) \to \mathrm{Mod_{el}}(\mathcal{C})$ iff $\mathsf{PB}f$ is an equivalence, iff f lies in the join of the classical and the Morita samenesses.*

Proof. By Lemma 4.2, $\mathrm{Mod_{el}}(f)$ is $\mathrm{Mod}(\mathsf{PB}f)$ for the coherent functor $\mathsf{PB}f$ between pretoposes. If $\mathrm{Mod}(\mathsf{PB}f)$ is an equivalence of categories then $\mathsf{PB}f$ is an equivalence, by the Makkai–Reyes conceptual completeness theorem [11, 7.1.8]; the converse is clear. The last clause is Theorem 4.1. □

Corollary 4.4. *For an interpretation $f\colon \mathcal{C} \to \mathcal{D}$ of coherent theories: f induces an equivalence of the categories of models with homomorphisms iff f is a Morita equivalence* [18, Thm. 6.1]*; f induces an equivalence of the categories of models with elementary embeddings iff f is a Morita equivalence up to Morleyization, that is $f \in V_{\mathsf{B}} \vee V_{\mathsf{P}}$. The second class is strictly larger: the inclusion of Example 3.5(i) induces an equivalence of the categories of models with elementary embeddings (two models up to isomorphism on each side, p true or false, and no elementary embedding between them, p being a sentence) and is not a Morita equivalence.*

Remark 4.5. For classical theories B is the identity and the two statements of Corollary 4.4 coincide, homomorphisms of models of a Boolean coherent theory being elementary embeddings; this is the setting of Hudetz's definable categorical equivalence [8] and of [18, Cor. 6.2]. The corollary says what changes when the theories are merely coherent: the category of models with elementary embeddings forgets the coherent content: the functor induced by an interpretation is an equivalence exactly when the interpretation is inverted by the Boolean pretopos completion. This concerns the functors induced by interpretations; an abstract equivalence of categories of models does not recover the theory [7].

5. The order is forced

Lemma 5.1. *For every coherent $\mathcal{C}$ the canonical functor $\mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$ is full and faithful, and the following are equivalent: it is an equivalence; $\mathsf{BP}\mathcal{C}$ is a pretopos; every equivalence relation of $\mathsf{BP}\mathcal{C}$ on an object of $\mathsf{P}\mathcal{C}$ has an effective quotient (is a kernel pair) in $\mathsf{BP}\mathcal{C}$.*

Proof. $\mathsf{PBP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$ (Proposition 3.4) and the unit $\mathsf{BP}\mathcal{C} \to \mathsf{PBP}\mathcal{C}$ of P is full and faithful [9, A3.3]. If $\mathsf{BP}\mathcal{C}$ is a pretopos the unit is an equivalence. If every equivalence relation on an object of $\mathsf{P}\mathcal{C}$ has a quotient: $\mathsf{BP}\mathcal{C}$ has

finite coproducts, since $\mathsf{P}\mathcal{C}$ has them and a subobject of a coproduct in the Boolean category is a coproduct of subobjects of the summands (first-order formulas over a coproduct sort reduce to the summands), and an equivalence relation on an arbitrary object $S \subseteq X$, $X \in \mathsf{P}\mathcal{C}$, extends by the diagonal to one on X, whose quotient contains S/E as the image of S; so $\mathsf{BP}\mathcal{C}$ is a pretopos. The remaining implication is trivial. □

The objects of $\mathsf{BP}\mathcal{C}$ are the first-order definable subobjects S of objects Y of $\mathsf{P}\mathcal{C}$, and $Y = (\coprod_{i \le r} D_i)/E$ with $D_i \in \mathcal{C}$ and E an equivalence relation of $\mathsf{F}\mathcal{C}$ [18, Thm. 4.4], so $E = \coprod_{i,j} E_{ij}$ with $E_{ij} \subseteq D_i \times D_j$ coherent: the *classical subobjects of the coherent imaginaries*. The objects of $\mathsf{PB}\mathcal{C}$ are the quotients of coproducts of classical definable sets by classical equivalence relations: the *classical imaginaries*.

Proposition 5.2. $\mathsf{BP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$ *when $\mathcal{C}$ is Boolean, and when $\mathcal{C}$ is a distributive lattice (a coherent propositional theory).*

Proof. If $\mathcal{C}$ is Boolean, $\mathsf{B}\mathcal{C} = \mathcal{C}$ and $\mathsf{P}\mathcal{C}$ is Boolean (Proposition 3.4), so both sides are $\mathsf{P}\mathcal{C}$. Let $\mathcal{C} = L$ be a distributive lattice. $\mathsf{B}L$ is the Boolean envelope $B(L)$, whose Stone space is the patch space of the spectral space $\operatorname{Spec} L$. The objects of $\mathsf{P}B(L)$, the coherent objects of $\operatorname{Sh}(\operatorname{Spec} B(L))$, are the finite coproducts $U_1 + \cdots + U_n$ of clopen subterminals: a coherent sheaf on a Stone space is covered by finitely many local sections over clopens, the locus where two of them agree is clopen, and refining to a finite clopen partition on which the sections are pairwise equal or disjoint gives the decomposition; each U_i is a Boolean combination of compact opens of $\operatorname{Spec} L$, an element of $B(L) = \mathsf{B}L \subseteq \mathsf{BP}L$, and $U_1 + \cdots + U_n$ is the subobject of the object $1 + \cdots + 1$ of $\mathsf{P}L$ cut out by the first-order formula "in the i-th summand and U_i". So $\mathsf{BP}L \to \mathsf{PB}L$ is essentially surjective, hence an equivalence by Lemma 5.1. □

Theorem 5.3 (the order is forced)**.** *Let T be the coherent theory of complete multipartite graphs: one sort σ, a relation $R \subseteq \sigma \times \sigma$ and the axioms*

$$R(x,x) \vdash \bot, \qquad R(x,y) \vdash R(y,x), \qquad R(x,z) \vdash R(x,y) \vee R(y,z),$$

whose models are the graphs whose non-adjacency is an equivalence relation, the complete multipartite graphs with any number of parts of any sizes. Let $\Pi := \sigma/\neg R$ be the quotient in $\mathsf{PB}\mathcal{C}_T$ of σ by the classical equivalence relation $\neg R(x,y)$, the set of parts*. Then Π is not isomorphic to any object of $\mathsf{BP}\mathcal{C}_T$. Hence $\mathsf{BP}\mathcal{C}_T$ is not a pretopos, $\mathsf{BP} \not\simeq \mathsf{PB}$, and the alternation of Theorem 4.1 cannot be performed with the completion first.*

Proof. $\neg R$ is an equivalence relation in $\mathsf{B}\mathcal{C}_T$: reflexive by the first axiom, symmetric by the second, and transitive since the third axiom is the contrapositive of $\neg R(x,y) \wedge \neg R(y,z) \vdash \neg R(x,z)$. An isomorphism $S \cong \Pi$ in $\mathsf{PB}\mathcal{C}_T$ gives, for every model M, a bijection $S(M) \cong \Pi(M)$ natural in elementary embeddings, in particular equivariant for $\operatorname{Aut}(M)$, where $\Pi(M)$ is the set of parts of M. Suppose $S \subseteq Y = (\coprod_{i \le r} D_i)/E$ is an object of $\mathsf{BP}\mathcal{C}_T$ as above,

$D_i \subseteq \sigma^{m_i}$. Each E_{ii} is a coherent formula in $2m_i$ variables, a disjunction of formulas $\exists \bar{z}\, \psi$ with ψ a conjunction of atoms $R(u,v)$ and $u = v$; let w bound the lengths of the $\bar{z}$ and m the m_i.

Let M_k be the complete k-partite graph with parts $P_1, \dots, P_k$ of size n, with $n \geq \max(1, 2m)$ and $k \geq 2m + w + 2$. Its automorphism group contains $S_n^k \rtimes S_k$, acting on $\Pi(M_k)$ through S_k. A point of $S(M_k)$ is a class $[\bar{d}]$ with $\bar{d} \in D_i(M_k)$ for some i; let $\mathcal{T}$ be the set of parts met by the coordinates of $\bar{d}$, $|\mathcal{T}| \leq m$. By the equivariant bijection the point corresponds to a part P_0, so it is fixed by S_n^k and moved by every transposition $\varepsilon = (P_0\, Q)$ of parts, $Q \neq P_0$. Choose $Q \notin \mathcal{T}$, $Q \neq P_0$, possible as $k > m + 1$. If $P_0 \notin \mathcal{T}$ then $\varepsilon \bar{d} = \bar{d}$ and the point is fixed by ε, a contradiction. So $P_0 \in \mathcal{T}$.

Let $g \in S_n^k$ move every coordinate of $\bar{d}$ to a vertex of the same part not occurring in $\bar{d}$ (possible as $n \geq 2m$). Since the point is fixed by g, $E_{ii}(\bar{d}, g\bar{d})$ holds in M_k: some disjunct $\exists \bar{z}\, \psi(\bar{x}, \bar{y}, \bar{z})$ holds at $\bar{x} = \bar{d}$, $\bar{y} = g\bar{d}$ with witnesses $\bar{c}$. We show that it holds at $\bar{x} = \bar{d}$, $\bar{y} = \varepsilon\bar{d}$ with witnesses $\bar{c}'$ defined as follows: if $c_l = d_s$ put $c'_l = d_s$; if $c_l = g d_t$ put $c'_l = \varepsilon d_t$ (the two cases are exclusive, $g\bar{d}$ sharing no vertex with $\bar{d}$); otherwise put c'_l a vertex in a part outside $\mathcal{T} \cup \{Q\}$, one new part and one vertex for each remaining value of $\bar{c}$, which $k \geq 2m + w + 2$ allows. Equalities: $d_s = d_{s'}$ is unchanged; $g d_t = g d_{t'}$ iff $d_t = d_{t'}$ iff $\varepsilon d_t = \varepsilon d_{t'}$; no atom $x_s = y_t$ occurs in ψ since $d_s \neq g d_t$; equalities involving $\bar{z}$ hold by construction. Atoms $R(u,v)$: in M_k, $R(a,b)$ holds iff a, b lie in different parts. The parts of $\bar{d}$ are unchanged; a coordinate $g d_t$ in part P is replaced by εd_t in part $\varepsilon(P)$; new witnesses lie in parts of their own. An atom $R(d_s, g d_t)$ in ψ holds, so $P := \mathrm{part}(d_t) \neq \mathrm{part}(d_s)$, and $\varepsilon(P)$ is P, or $Q \notin \mathcal{T}$ when $P = P_0$; either way $\varepsilon(P) \neq \mathrm{part}(d_s)$. An atom $R(g d_t, g d_{t'})$ holds iff $\mathrm{part}(d_t) \neq \mathrm{part}(d_{t'})$, preserved by the bijection ε of parts. Atoms with a new witness hold since its part is new, and atoms with a witness copied from a coordinate reduce to the previous cases. Hence $E_{ii}(\bar{d}, \varepsilon\bar{d})$, so the point $[\bar{d}]$ is fixed by ε, which moves P_0: a contradiction. Thus $S(M_k)$ is empty, while $\Pi(M_k)$ has k points. The last claims follow from Lemma 5.1. □

Remark 5.4 (uniformity)**.** For each $k \geq 1$ the coherent formula

$$E_k(x,y) \;:=\; x = y \;\vee\; \exists z_1 \dots z_{k-1} \Big(\bigwedge_j \big(R(x, z_j) \wedge R(y, z_j) \big) \wedge \bigwedge_{j<l} R(z_j, z_l) \Big)$$

is an equivalence relation in T: in a model with k' parts, x and y have $k-1$ pairwise adjacent common neighbours iff $k-1$ parts other than theirs exist, so E_k is equality if $k' < k$, the relation "same part" if $k' = k$ and total if $k' > k$. Hence σ/E_k is a coherent imaginary which agrees with Π on the k-partite models, and the object [exactly k parts] $\times\, \sigma/E_k$ of BPC_T agrees with Π on the k-partite models and is empty elsewhere. The theorem says that no single object of BPC_T agrees with Π on all models, not even on all models with finitely many parts (the theory also has models with infinitely many parts, on which every sentence "exactly k parts" fails while

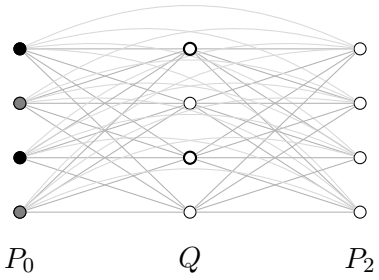


FIGURE 1. M_3 with $n = 4$. A pair $\bar{d}$ in P_0 (black), its image $g\bar{d}$ under a permutation of P_0 (grey) and its image $\varepsilon\bar{d}$ under the transposition $(P_0\, Q)$ (white). For k large with respect to the formula, a coherent relation between $\bar{d}$ and $g\bar{d}$ witnessed in M_k is also witnessed between $\bar{d}$ and $\varepsilon\bar{d}$, since positive atoms only ask for distinct parts.

Π is nonempty): the number of witnesses $k-1$ is unbounded, and a single coherent equivalence relation, with w witnesses, cannot see the parts of a k-partite model for $k > 2m + w + 2$. (The proof of the theorem is this direct count, not an appeal to the compactness theorem, which enters in Theorem 6.1.) The obstruction is the same finiteness of formulas that makes the sharp object of [18, Thm. 5.6] unreachable in three Morita steps.

Corollary 5.5 (lengths of the alternation)**.** *For every coherent $\mathcal{C}$ the alternation starting with the completion stops after three steps, $\mathsf{PBP}\mathcal{C} \simeq \mathsf{PB}\mathcal{C}$, and the one starting with the translation after two. The three steps are needed for the theory of complete multipartite graphs, and two suffice for Boolean and for propositional theories.*

Proof. Proposition 3.4, Theorem 5.3 and Proposition 5.2. □

$$\begin{array}{ccccc} \mathcal{C} & \xrightarrow{\eta_{\mathsf{B}}} & \mathsf{B}\mathcal{C} & \xrightarrow{\eta_{\mathsf{P}}} & \mathsf{PB}\mathcal{C} \\ {\scriptstyle \eta_{\mathsf{P}}}\downarrow & & & & \uparrow{\scriptstyle \simeq} \\ \mathsf{P}\mathcal{C} & \xrightarrow[\eta_{\mathsf{B}}]{} & \mathsf{BP}\mathcal{C} & \xrightarrow[\eta_{\mathsf{P}}]{} & \mathsf{PBP}\mathcal{C} \end{array}$$

The upper row reaches the join in two steps; in the lower row the last unit is not an equivalence for the theory of Theorem 5.3.

Remark 5.6. Lemma 5.1 places the question in classical model theory: $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ iff the Morleyization T^m eliminates imaginaries relative to the coherent imaginaries of T, that is, iff every T^m-definable equivalence relation on a coherent imaginary has a quotient among the classical subobjects of coherent imaginaries. Theorem 5.3 is a failure of this relative elimination, by a count depending on the size of the candidate formula; Proposition 5.2 is its success for propositional theories, where imaginaries are coproducts of subterminals. The next section characterises the relative elimination.

6. A Galois characterisation of the commutation

A *coherent imaginary* of T is an object $Y = (\coprod_i D_i)/F$ of $\mathsf{P}\mathcal{C}_T$, and a *classical imaginary* is an object X/E of $\mathsf{PB}\mathcal{C}_T$, X a classical definable set and E a classical equivalence relation. Call X/E *coherently representable* if it is isomorphic in $\mathsf{PB}\mathcal{C}_T$ to a classical subobject of a coherent imaginary, i.e. to an object of $\mathsf{BP}\mathcal{C}_T$. Representable objects are closed under classical subobjects, and every classical imaginary is a classical subobject of one with $X \in \mathsf{F}\mathcal{C}_T$: a classical definable set X is a subobject of a finite coproduct P of products of sorts, $E \cup \Delta_P$ is a classical equivalence relation on P, and X/E is the image of X in $P/(E \cup \Delta_P)$; an equivalence relation on a quotient of P pulls back to one on P with the same quotient. So $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ iff every X/E with $X \in \mathsf{F}\mathcal{C}_T$ is coherently representable (Lemma 5.1). For a model M and a point c of an imaginary at M, $\mathrm{Aut}(M)_c$ is its stabiliser.

Theorem 6.1 (Galois criterion)**.** *For a coherent theory T and a classical imaginary X/E the following are equivalent.*

(i) *X/E is coherently representable.*
(ii) *For every model M of T and every $c \in (X/E)(M)$ there are a coherent imaginary Y of T and $y \in Y(M)$ with $\mathrm{Aut}(M)_c = \mathrm{Aut}(M)_y$.*
(iii) *The same for the κ-saturated, strongly κ-homogeneous models of the completions of T^m (the complete consistent classical theories extending it, those with finite models included), for some cardinal κ greater than the cardinality of the language of T^m (which is that of the language of T plus $\aleph_0$); such models exist for every κ.*

Hence $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$ iff every classical imaginary satisfies (ii), iff every classical imaginary satisfies (iii).

Proof. (i)⇒(ii): an isomorphism $X/E \cong S \subseteq Y$ gives bijections $(X/E)(M) \cong S(M) \subseteq Y(M)$ natural in elementary embeddings, in particular $\mathrm{Aut}(M)$-equivariant, so c corresponds to a point y with the same stabiliser. (ii)⇒(iii) is trivial. (iii)⇒(i): fix a completion T' of T^m and a model $\mathfrak{M}$ as in (iii); we work in the many-sorted structure $\mathfrak{M}^{\mathsf{P}}$ whose sorts are the objects of $\mathsf{PB}\mathcal{C}_T$ evaluated at $\mathfrak{M}$, that is $\mathfrak{M}^{\mathrm{eq}}$ with finite coproducts of sorts added, and write $\mathrm{dcl}^{\mathrm{eq}}$, $\mathrm{acl}^{\mathrm{eq}}$ in it. $\mathfrak{M}$ and $\mathfrak{M}^{\mathsf{P}}$ have the same automorphisms, and $\mathfrak{M}^{\mathsf{P}}$ is again κ-saturated and strongly κ-homogeneous: an element of a quotient of a coproduct is the class of a tuple of $\mathfrak{M}$ in one of finitely many summands, so types and partial elementary maps of $\mathfrak{M}^{\mathsf{P}}$ lift to $\mathfrak{M}$. Let $c = [a]_E$ and $y \in Y(\mathfrak{M})$ have the same stabiliser. By strong homogeneity each of c, y is in the definable closure of the other over $\emptyset$, by the standard fact that in a κ-saturated strongly κ-homogeneous model the definable closure of a set A of fewer than κ imaginaries is the set of imaginaries fixed by every automorphism over A [13, Ch. 16]: if $y \notin \mathrm{dcl}^{\mathrm{eq}}(c)$ then $\mathrm{tp}(y/c)$ has a second realisation by saturation, and an automorphism over c moves y to it by homogeneity. So there are $\emptyset$-definable partial maps $f\colon X/E \rightharpoonup Y$, $g\colon Y \rightharpoonup X/E$ with $f(c) = y$, $g(y) = c$. The

set $U := \{x \in X : f([x]_E) \text{ is defined and } gf([x]_E) = [x]_E\}$ is $\emptyset$-definable, E-saturated, contains the class c, and f is injective on U/E. The sets U so obtained, one for each class c, cover $X(\mathfrak{M})$, and finitely many of them cover X: otherwise the partial type $\{x \in X\} \cup \{x \notin U : U \text{ as above}\}$, which has fewer than κ formulas, κ exceeding the cardinality of the language, is finitely satisfiable, hence realised in $\mathfrak{M}$ by some a' (the formulas are over $\emptyset$ and fewer than κ, since κ exceeds the size of the language), which lies in the set U attached to its own class. Let $U_1, \dots, U_N$ cover X, with maps f_j into coherent imaginaries Y_j of T. Then $\theta\colon x \mapsto f_j([x]_E)$ for $x \in U_j \setminus \bigcup_{i<j} U_i$ is a definable map $X \to \coprod_j Y_j$, total since the U_j cover X, and with kernel E: $\theta(x) = \theta(x')$ forces the same j and $f_j([x]_E) = f_j([x']_E)$, hence $[x]_E = [x']_E$ by injectivity. So X/E is isomorphic to the image S of θ, a classical subobject of the coherent imaginary $\coprod_j Y_j$. The statement "θ is total with kernel E" is a sentence $\varphi_{T'}$ of T^m true in T'. Every completion of T^m contains one of these sentences, so by compactness $T^m \vdash \varphi_1 \vee \dots \vee \varphi_K$ for finitely many of them, with maps θ_k and images S_k; then $X \to \coprod_k [\varphi_k \wedge \neg\varphi_1 \wedge \dots \wedge \neg\varphi_{k-1}] \times S_k$, $x \mapsto \theta_k(x)$ in the k-th summand for the first k with φ_k true, is total with kernel E in every model of T^m, so X/E is isomorphic to its image, a classical subobject of a coherent imaginary. The last sentence is Lemma 5.1 and the reduction to $X \in \mathsf{FC}_T$ above. □

Theorem 6.1 is Poizat's criterion [12], that a theory eliminates imaginaries iff in a saturated model every imaginary has the stabiliser of a real tuple, with real tuples replaced by points of coherent imaginaries; imaginaries defined by positive existential formulas are related to coherent functors in [15]. The next proposition turns it into a test that needs no saturation. The *coherent type* $\mathrm{ctp}(\bar{a})$ of a tuple in M is the set of coherent formulas it satisfies.

Proposition 6.2 (coherent-type criterion)**.** *Let M be a model of T, X/E a classical imaginary and $c \in (X/E)(M)$. Suppose that for every finite tuple $\bar{d}$ of elements of M there are $g \in \mathrm{Aut}(M)_c$ and $\varepsilon \in \mathrm{Aut}(M) \setminus \mathrm{Aut}(M)_c$ with $\mathrm{ctp}(\bar{d}, g\bar{d}) \subseteq \mathrm{ctp}(\bar{d}, \varepsilon\bar{d})$. Then X/E is not coherently representable, and $\mathsf{BPC}_T \not\simeq \mathsf{PBC}_T$.*

Proof. Otherwise, by Theorem 6.1(ii), c has the stabiliser of a point $y = [\bar{d}]_F$ of a coherent imaginary $(\coprod D_i)/F$, with $\bar{d} \in D_i(M)$. As g fixes c it fixes y, so $F_{ii}(\bar{d}, g\bar{d})$ holds, and $F_{ii} \in \mathrm{ctp}(\bar{d}, g\bar{d}) \subseteq \mathrm{ctp}(\bar{d}, \varepsilon\bar{d})$; thus $F_{ii}(\bar{d}, \varepsilon\bar{d})$, ε fixes y and hence c, a contradiction. □

The inclusion of coherent types is verified by local homomorphisms: $\mathrm{ctp}(\bar{d}, \bar{b}) \subseteq \mathrm{ctp}(\bar{d}, \bar{b}')$ as soon as for every finite set $W \supseteq \bar{d}\bar{b}$ of elements of M there is a map $h\colon W \to M$ which preserves the positive atoms true in W and has $h|_{\bar{d}} = \mathrm{id}$, $h(\bar{b}) = \bar{b}'$, since the witnesses of a coherent formula lie in such a W.

Example 6.3 (Theorem 5.3 in one model)**.** Let M_∞ be the complete multipartite graph with countably many countably infinite parts, so that every

permutation of the parts is induced by an automorphism, and c a part P_0. For a tuple $\bar{d}$ meeting the parts $\mathcal{T}$: if $P_0 \notin \mathcal{T}$ take $g = \mathrm{id}$ and $\varepsilon = (P_0\, Q)$ with $Q \notin \mathcal{T}$, $Q \neq P_0$, so that $\varepsilon\bar{d} = \bar{d}$; if $P_0 \in \mathcal{T}$ take g moving each coordinate to a fresh vertex of its part and $\varepsilon = (P_0\, Q)$, $Q \notin \mathcal{T}$, and the local homomorphism h fixing $\bar{d}$, sending gd_t to εd_t and the other elements of W to distinct fresh parts, which preserves equalities and the positive atoms $R(u, v)$ exactly as in the proof of Theorem 5.3, now with no bound on the number of parts. Proposition 6.2 gives $\Pi \notin \mathsf{BPC}_T$. The finite models M_k of Theorem 5.3 show in addition that the obstruction is visible in finite models, with k depending on the size of the candidate.

Corollary 6.4 (weak elimination of imaginaries)**.** *If every completion of the Morleyization T^m has weak elimination of imaginaries, i.e. for every imaginary c in a saturated model there is a real tuple $\bar{a}$ with $c \in \mathrm{dcl}^{\mathrm{eq}}(\bar{a})$ and $\bar{a} \in \mathrm{acl}^{\mathrm{eq}}(c)$, then $\mathsf{BPC}_T \simeq \mathsf{PBC}_T$. This covers the theories whose Morleyization eliminates imaginaries, and the theories of infinite sets and of dense linear orders without endpoints* [4].

Proof. Weak elimination is assumed for the ordinary imaginaries, the classes of tuples under definable equivalence relations. A point $c = [a]_E$ of a classical imaginary X/E, X a classical subobject of a coproduct $\coprod_i P_i$ of products of sorts with a in the summand P_i, is interdefinable with the ordinary imaginary $c \cap P_i$, the class of a under the restriction of E to $X \cap P_i$; so it suffices to treat ordinary imaginaries. Let A be the orbit of $\bar{a}$ under $\mathrm{Aut}(\mathfrak{M})_c$, a finite set of n tuples in a product of sorts P, as $\bar{a} \in \mathrm{acl}^{\mathrm{eq}}(c)$. An automorphism fixing c permutes A; one fixing A setwise sends $\bar{a}$ to some $h\bar{a}$ with h fixing c, hence fixes $c = f(\bar{a})$, f a $\emptyset$-definable partial function with $f(\bar{a}) = c$, defined on the orbit of $\bar{a}$. So $\mathrm{Aut}(\mathfrak{M})_c = \mathrm{Aut}(\mathfrak{M})_A$. The finite set A is a point of the classical subobject "pairwise distinct" of the coherent imaginary P^n/S_n, the quotient by the coherent equivalence relation $\bigvee_{\pi \in S_n} \bar{y} = \pi\bar{x}$. Apply Theorem 6.1. □

Example 6.5 (a non-Boolean case)**.** The complete theories named in Corollary 6.4 have Boolean syntactic categories, for which the commutation is already Proposition 5.2; the corollary has content for coherent theories that are not Boolean. Let $T_=$ be the empty coherent theory on one sort. $\mathcal{C}_{T_=}$ is not Boolean: $x \neq y$ is not equivalent to a coherent formula, coherent formulas being preserved by the homomorphism collapsing a two-element model to a point. The completions of $T_=^m$ are the theory of infinite sets, which has weak elimination of imaginaries [4], and the theories of the sets with $n \geq 0$ elements, which have it as every finite structure does: for $\bar{a}$ an enumeration of the structure, $\bar{a} \in \mathrm{acl}^{\mathrm{eq}}(c)$ since the structure is finite, and $c \in \mathrm{dcl}^{\mathrm{eq}}(\bar{a})$ since only the identity fixes $\bar{a}$. Hence $\mathsf{BPC}_{T_=} \simeq \mathsf{PBC}_{T_=}$: every classical imaginary of pure equality is a classical subobject of a coherent one, although the two categories $\mathcal{C}_{T_=}$ and $\mathsf{BC}_{T_=}$ differ.

Remark 6.6. The sign of the negation decides the two basic cases. In the coherent theory of an equivalence relation E the classes are the points of the coherent imaginary σ/E. In the theory of complete multipartite graphs the classes of the classical equivalence relation $\neg R$ are not coherently representable (Example 6.3), although $\neg R$ is a Boolean combination of coherent formulas: the relative elimination of imaginaries is not decided by the definability of the equivalence relation but by the stabilisers of its classes.

7. Free theories: a dichotomy

The two criteria decide the commutation for the coherent theories without axioms.

Theorem 7.1 (dichotomy)**.** *Let Σ be a relational signature, with any set of sorts and relation symbols of any finite arities, and T_Σ the coherent theory over Σ with no axioms. Then $\mathsf{BP}\mathcal{C}_{T_\Sigma} \simeq \mathsf{PB}\mathcal{C}_{T_\Sigma}$ iff Σ is monadic, that is, every relation symbol has arity at most one.*

Lemma 7.2 (supports)**.** *Let Ω be an infinite set and $F_1, F_2 \subseteq \Omega$ finite. The pointwise stabilisers of F_1 and of F_2 in $\mathrm{Sym}(\Omega)$ generate the pointwise stabiliser of $F_1 \cap F_2$.*

Proof. Write $\mathrm{Fix}(F)$ for the pointwise stabiliser, $F = F_1 \cap F_2$, $n = |F_1 \setminus F|$, and H for the generated group. For $k \in \mathrm{Fix}(F_2)$, $k\,\mathrm{Fix}(F_1)k^{-1} = \mathrm{Fix}(kF_1) \subseteq H$, and kF_1 can be $F \cup D$ for any n-element D disjoint from F_2. Let $g \in \mathrm{Fix}(F)$. Choose such a D, then such a D' disjoint from $D \cup g(D)$, and $p \in \mathrm{Fix}(F \cup D')$ with $p(g(d)) = d$ for $d \in D$, which exists since the partial injection $g(d) \mapsto d$ avoids $F \cup D'$ and Ω is infinite. Then $pg \in \mathrm{Fix}(F \cup D) \subseteq H$ and $p \in H$, so $g \in H$. □

Proof of Theorem 7.1. A symbol of arity at least two. Let $S \subseteq \sigma_1 \times \sigma_2 \times \cdots$ have arity at least two (σ_1, σ_2 possibly equal). Let M be the Σ-structure in which every sort is a copy of the vertex set of M_∞ (Example 6.3), the parts being indexed by the same set in every sort, $S(u_1, u_2, \dots)$ holds iff u_1 and u_2 lie in parts of different indices, and every other relation symbol is empty. The formula $E(x, y) := \forall \bar{z}\,(S(x, \bar{z}) \leftrightarrow S(y, \bar{z}))$ is an equivalence relation on σ_1 in every Σ-structure, and in M its classes are the parts of σ_1. Let c be a part P_0. The automorphisms of M include the permutations of the index set, acting on all sorts at once, and independent permutations inside each part of each sort. For a tuple $\bar{d}$ take g, $\varepsilon = (P_0\, Q)$ and the local homomorphism h as in Example 6.3: the positive atoms true in M are equalities and S-atoms, the latter holding iff the first two arguments have different indices, and they are preserved as the atoms $R(u, v)$ were. By Proposition 6.2, σ_1/E is not coherently representable.

Monadic Σ. By Corollary 6.4 it suffices that every completion of T^m_Σ has weak elimination of imaginaries. This is known [2, Rem. 7.14]; we include the short argument. An imaginary $c = [\bar{a}]_E$ of a saturated model M involves a finite sublanguage; the reduct of M to it is saturated, and membership in

$\mathrm{dcl}^{\mathrm{eq}}$ and $\mathrm{acl}^{\mathrm{eq}}$ established in the reduct persists in M. So let Σ be finite. Then M is the disjoint union of finitely many sets $A_1, \ldots, A_r$, the nonempty Boolean atoms of the predicates in each sort, each finite or infinite, and $\mathrm{Aut}(M) = \prod_i \mathrm{Sym}(A_i)$. Let B_0 be the union of the finite A_i. Call a finite $B \supseteq B_0$ a *support* of c if every automorphism fixing B pointwise fixes c; the union of B_0 and the coordinates of $\bar{a}$ is one. By Lemma 7.2, applied in each infinite A_i, the intersection of two supports is a support, so c has a least support B. If g fixes c then gB is a support, so $B \subseteq gB$ and $gB = B$. Hence an enumeration $\bar{b}$ of B has a finite orbit under $\mathrm{Aut}(M)_c$, so $\bar{b} \in \mathrm{acl}^{\mathrm{eq}}(c)$, and c is fixed by the automorphisms fixing $\bar{b}$, so $c \in \mathrm{dcl}^{\mathrm{eq}}(\bar{b})$. □

Remark 7.3 (monotonicity)**.** Proposition 6.2 uses the model and the coherent formulas but not the axioms. So a failure which it witnesses at a model of T' holds for every coherent theory $T \subseteq T'$ in the same signature in which E is still provably an equivalence relation. The first half of the proof above is Theorem 5.3 read in this way: on the complete multipartite graphs the relation "same neighbourhood", an equivalence relation without any axiom, is $\neg R$.

Remark 7.4. The monadic theories without axioms are not Boolean when they have a sort: $x \neq y$ and, for a unary predicate P, $\neg P(x)$ are not coherent. So the commutation of the classical translation with the Morita completion is not a Boolean phenomenon, and for free theories its failure has a single cause, a relation symbol of arity at least two, whose classes of "same neighbourhood" are classical imaginaries which no coherent quotient sees. For theories with axioms Theorem 6.1 gives the criterion; the next section applies it to the coherent theory of an equivalence relation.

8. An equivalence relation: commutation is not a classical invariant

Corollary 8.1 (relative weak elimination)**.** *Suppose that in the saturated strongly homogeneous models of the completions of T^m, as in Theorem 6.1(iii), every ordinary classical imaginary c has a finite tuple $\bar{b}$ of points of coherent imaginaries of T with $c \in \mathrm{dcl}^{\mathrm{eq}}(\bar{b})$ and $\bar{b} \in \mathrm{acl}^{\mathrm{eq}}(c)$. Then $\mathsf{BP}\mathcal{C}_T \simeq \mathsf{PB}\mathcal{C}_T$.*

Proof. The proof of Corollary 6.4 with P a product of coherent imaginaries, an object of $\mathsf{P}\mathcal{C}_T$: the orbit of $\bar{b}$ under the stabiliser of c is a point of a classical subobject of P^n/S_n, again a coherent imaginary, with the stabiliser of c. □

Theorem 8.2. *Let T_E be the coherent theory of an equivalence relation E on one sort σ. Then $\mathsf{BP}\mathcal{C}_{T_E} \simeq \mathsf{PB}\mathcal{C}_{T_E}$.*

Proof. Consider first a completion whose models are finite, and its model M. A tuple $\bar{b}$ enumerating M has a finite orbit and every automorphism fixing $\bar{b}$ is the identity, so every imaginary c is fixed by the automorphisms

fixing $\bar{b}$ and $\bar{b}$ has a finite orbit under the stabiliser of c; Corollary 8.1 applies to this completion. For the other completions, whose models are infinite, let M be a saturated model, of uncountable cardinality κ (these completions are ω-stable), $Q = \sigma/E$ the coherent imaginary of classes, and I_k the set of classes with k elements, $1 \leq k \leq \infty$; each I_k is finite or of cardinality κ, and the infinite classes have κ elements. Every bijection of σ preserving E is an automorphism, so $\mathrm{Aut}(M) = \prod_k G_k$ with G_k the group of such bijections of the union of the classes of size k. Fix an imaginary c, a representative $\bar{a}$ of it, and the finite set J of the sizes of the classes of the coordinates of $\bar{a}$; the product of the factors G_k, $k \notin J$, fixes $\bar{a}$, hence c, and every element or class of a size in J. (Infinitely many of the sets I_k may be finite and nonempty, which is why the argument is restricted to J.) Let A_0 consist of the classes in the finite sets I_j, $j \in J$, and of the elements of the finite classes among them; it is finite. Call a finite set $B \subseteq \sigma \sqcup Q$ of elements and classes of sizes in J *closed* if $A_0 \subseteq B$, the class of every element of B is in B, and every finite class in B has all its elements in B; closed sets are stable under intersection and under automorphisms. A closed B is a *support* of c if every automorphism fixing B pointwise fixes c; the closure of A_0 and the coordinates of $\bar{a}$ is one.

We claim that for closed B_1, B_2 the pointwise stabilisers $\mathrm{Fix}(B_1)$, $\mathrm{Fix}(B_2)$ generate $\mathrm{Fix}(B)$, $B = B_1 \cap B_2$. Let H be the generated group; it contains the factors G_k, $k \notin J$, which fix B_1, so it suffices to show that H contains every $g \in \mathrm{Fix}(B)$ which is the identity outside the classes of sizes in J. For $k \in \mathrm{Fix}(B_2)$, $\mathrm{Fix}(kB_1) \subseteq H$ and $kB_1 = B \cup D$ with $D = k(B_1 \setminus B)$. The points of $B_1 \setminus B$ are of two kinds: classes outside B_2, which belong to infinite sets I_j since $A_0 \subseteq B$, with their elements listed in B_1, and k may send them to any classes of the same size outside B_2; and elements whose class is in B_2, hence in B, the class being infinite since finite classes of B_2 lie in B_2, and k may send them to any elements of that class outside B_2. Let g be as above. Choose such a D, and then such a D' disjoint from $D \cup g(D)$ whose classes outside B are different from the classes of the points of $D \cup g(D)$. The map sending $g(d)$ to d for $d \in D$ and fixing $B \cup D'$ respects E and the sizes of classes, as g is an automorphism fixing B and finite classes occur in D with all their elements; it extends to an automorphism p: infinitely many classes of each size concerned are left, infinitely many elements of each infinite class concerned are left, and the finite classes concerned are mapped with all their elements. Then $pg \in \mathrm{Fix}(B \cup D) \subseteq H$ and $p \in \mathrm{Fix}(B \cup D') \subseteq H$, so $g \in H$.

Hence the intersection of two supports is a support and c has a least support B; an automorphism fixing c maps B to a support, hence onto B. So an enumeration $\bar{b}$ of B, a tuple of points of σ and Q, has a finite orbit under the stabiliser of c, and c is fixed by the automorphisms fixing $\bar{b}$. By Corollary 8.1 the orders agree. □

Corollary 8.3. *The commutation of the Morleyization with the pretopos completion is not invariant under the classical sameness: the theory T_E of*

an equivalence relation and the theory T of complete multipartite graphs of Theorem 5.3 have equivalent Morleyizations, by $R \leftrightarrow \neg E$, so that $\mathcal{C}_{T_E}$ and $\mathcal{C}_T$ are V_{B}-equivalent, and the two orders agree for the first and not for the second.

Proof. Under $R := \neg E$, $E := \neg R$ the axioms of each theory become theorems of the Morleyization of the other, so $\mathsf{B}\mathcal{C}_{T_E} \simeq \mathsf{B}\mathcal{C}_T$, and both receive the units from $\mathcal{C}_{T_E}$ and $\mathcal{C}_T$, which lie in V_{B}. The rest is Theorems 8.2 and 5.3. □

The classical theory is the same and so is the classical imaginary, the set of classes; it is a coherent imaginary when the relation named by the signature is the equivalence, and not coherently representable when it is its complement. What the coherent presentation adds to the classical theory is the choice of which definable relations are positive, and the commutation depends on this choice.

9. Excluded middle is not the obstruction

The Morleyization is a composite of two translations, the passage from coherent to intuitionistic first-order logic and the addition of excluded middle, and the forced order comes from the first.

Let **Heyt** be the 2-category of small Heyting categories, the coherent categories in which the pullback of subobjects along every morphism has a right adjoint $\forall_f$, with the coherent functors preserving these adjoints (Heyting functors) and natural isomorphisms [9, A1.4]; **BCoh** is a full sub-2-category of it. Let $\mathsf{H}\mathcal{C}$ be the syntactic category of the internal coherent theory of $\mathcal{C}$ read in intuitionistic first-order logic, and $\mathsf{K}\mathcal{D}$ that of the internal first-order theory of a Heyting category $\mathcal{D}$ with excluded middle added (the classicalisation; not to be confused with the double negation translation of Gödel and Gentzen, which goes from classical to intuitionistic logic). As in Proposition 3.2, a coherent functor from $\mathcal{C}$ to a Heyting category interprets every first-order formula and an isomorphism of such models preserves the interpretations, and a Heyting functor into a Boolean category satisfies excluded middle; so

$$\mathbf{Heyt}(\mathsf{H}\mathcal{C}, \mathcal{D}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{D}), \qquad \mathbf{Coh}(\mathsf{K}\mathcal{D}, \mathcal{B}) \simeq \mathbf{Heyt}(\mathcal{D}, \mathcal{B})$$

for $\mathcal{D}$ Heyting and $\mathcal{B}$ Boolean, and $\mathsf{B} \simeq \mathsf{KH}$.

Lemma 9.1. *If $\mathcal{D}$ is a Heyting category then $\mathsf{P}\mathcal{D}$ is one, the unit $\mathcal{D} \to \mathsf{P}\mathcal{D}$ is a Heyting functor, and the extension to $\mathsf{P}\mathcal{D}$ of a Heyting functor from $\mathcal{D}$ to a Heyting pretopos is a Heyting functor. Hence PK is a reflector of* **Heyt** *onto* **BPretop** *and $\mathsf{PKP} \simeq \mathsf{PK}$.*

Proof. An object of $\mathsf{P}\mathcal{D}$ is X/F with X a finite coproduct of objects of $\mathcal{D}$ and F an equivalence relation, and its subobjects are the F-saturated subobjects of X, a subobject of a coproduct being a family of subobjects of the summands. Saturated subobjects are closed under the Heyting implication

of X, and for a morphism $X/F \to Y/G$ with graph $\Gamma \subseteq X \times Y$ the subobject $\{y : \forall x\,(\Gamma(x,y) \Rightarrow S(x))\}$ of Y is G-saturated and is the right adjoint of the pullback at a saturated S. These formulas are preserved by the unit and by the extension of a Heyting functor. The last sentence follows as in Proposition 3.4. □

Theorem 9.2 (classicalisation commutes)**.** *For every Heyting category $\mathcal{D}$ the canonical functor $\mathsf{KP}\mathcal{D} \to \mathsf{PK}\mathcal{D}$ is an equivalence.*

Proof. It is the unit $\mathsf{KP}\mathcal{D} \to \mathsf{PKP}\mathcal{D}$ of P, full and faithful, followed by the equivalence of Lemma 9.1. For essential surjectivity let X/E be an object of $\mathsf{PK}\mathcal{D}$: X a finite coproduct of objects of $\mathcal{D}$, the objects of $\mathsf{K}\mathcal{D}$ being the first-order formulas over $\mathcal{D}$ as are those of $\mathcal{D}$, and E a first-order formula which the classical theory proves to be an equivalence relation on X. Put

$$F(x,y) \;:=\; \forall z\,\big(E(x,z) \leftrightarrow E(y,z)\big) \wedge \forall z\,\big(E(z,x) \leftrightarrow E(z,y)\big),$$

z ranging over X, summand by summand. In intuitionistic logic, without any axiom, F is an equivalence relation, the biconditional being reflexive, symmetric and transitive; so X/F is an object of $\mathsf{P}\mathcal{D}$. Classically F and E coincide: $F(x,y)$ gives $E(x,y)$ by $z = y$ and reflexivity, and $E(x,y)$ gives $F(x,y)$ by symmetry and transitivity. So the image of X/F in $\mathsf{PK}\mathcal{D}$ is X/E. □

Corollary 9.3. *On* **Heyt** *the join $V_{\mathsf{K}} \vee V_{\mathsf{P}} = V_{\mathsf{PK}}$ is reached by one alternation in either order. For a coherent category $\mathcal{C}$ the comparison $\mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$ is the image under K of the comparison $\mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$, followed by the equivalence $\mathsf{KPH}\mathcal{C} \simeq \mathsf{PKH}\mathcal{C}$. In particular, when the orders agree for H they agree for B, and for the theory of complete multipartite graphs and for the theories without axioms with a relation symbol of arity at least two the orders disagree already for H.*

Proof. Lemma 2.1 with Lemma 9.1 and Theorem 9.2; $\mathsf{B} = \mathsf{KH}$; and Theorems 5.3 and 7.1. □

The formula F, "same neighbourhood", is the witness on both sides. It is first-order and intuitionistically an equivalence relation, so that every classical imaginary is the classicalisation of a quotient defined intuitionistically. For some E it is coherent (for E the equality it is the equality); in Theorems 5.3 and 7.1 its quotient is not a classical subobject of any coherent imaginary, which is the obstruction there. The forced order of translation and completion thus arises in the Heyting completion, not in the addition of excluded middle.

10. Further translations

The reading of this paper applies to any translation which is a reflector, or more generally a 2-functor, on a 2-category of theories, and the three questions it answers for the Morleyization are to be asked for each: the join

with the Morita sameness and its alternation length; the category of models that detects the join; the commutation with the completions F, E, P.

(1) The Heyting completion H of Section 9: by Corollary 9.3 it carries the obstruction, and the question is whether the orders agree for H exactly when they agree for B, and what replaces the Galois criterion in intuitionistic logic, where models in **Set** no longer suffice.
(2) The Gödel–McKinsey–Tarski translation, which reads a Heyting algebra as the open elements of an interior operator on a Boolean algebra ($\mathsf{S4}$); joins of closure operators and their alternation lengths are the subject of [17], and the question is whether a translation of this kind defines a point of the lattice of samenesses.
(3) Maruyama's Lawvere–Tierney topologies on syntactic hyperdoctrines as translations [14, §4.10]: for an idempotent j, $\rho_j(T) = \{\varphi : T \vdash j\varphi\}$ is a closure operator on theories, and the question is which j commute with the Morita completion. The Morleyization is not of this form, since it enlarges the syntactic category; Theorem 5.3 answers the analogous question for it.

Declarations

The author has no competing interests to declare.

References

[1] T. W. Barrett, H. Halvorson, *Morita equivalence*, Rev. Symb. Log. 9 (2016), 556–582.
[2] G. Conant, C. d'Elbée, Y. Halevi, L. Jimenez, S. Rideau-Kikuchi, *Enriching a predicate and tame expansions of the integers*, J. Math. Log. 25 (2025), no. 1, 2450011.
[3] A. Carboni, E. M. Vitale, *Regular and exact completions*, J. Pure Appl. Algebra 125 (1998), 79–116.
[4] E. Casanovas, R. Farré, *Weak forms of elimination of imaginaries*, Math. Logic Quart. 50 (2004), 126–140.
[5] A. D'Arienzo, V. Pagano, I. M. J. McInnis, *Bicategories, biequivalence, and bi-interpretability*, preprint, arXiv:2011.14056v2 (2023); the numbering is that of this version.
[6] V. Harnik, *Model theory vs. categorical logic: two approaches to pretopos completion (a.k.a. T^{eq})*, in: Models, Logics, and Higher-Dimensional Categories, CRM Proc. Lecture Notes 53, Amer. Math. Soc., 2011, 79–106.
[7] H. Halvorson, D. Tsementzis, *Categories of scientific theories*, in: E. Landry (ed.), Categories for the Working Philosopher, Oxford University Press, 2017, 402–429.
[8] L. Hudetz, *Definable categorical equivalence*, Philos. Sci. 86 (2019), 47–75.
[9] P. T. Johnstone, *Sketches of an Elephant: A Topos Theory Compendium*, Oxford Univ. Press, 2002.
[10] S. Lack, E. M. Vitale, *When do completion processes give rise to extensive categories?*, J. Pure Appl. Algebra 159 (2001), 203–230.
[11] M. Makkai, G. E. Reyes, *First Order Categorical Logic*, Lecture Notes in Math. 611, Springer, 1977.
[12] B. Poizat, *Une théorie de Galois imaginaire*, J. Symbolic Logic 48 (1983), 1151–1170.
[13] B. Poizat, *A Course in Model Theory*, Universitext, Springer, 2000.
[14] Y. Maruyama, *Meaning and Duality: From Categorical Logic to Quantum Physics*, D.Phil. thesis, University of Oxford, 2016.

[15] R. Rajani, M. Prest, *Model-theoretic imaginaries and coherent sheaves*, Appl. Categ. Structures 17 (2009), 517–559.

[16] R. Shibusawa, *The sameness lattice of a category: weak-equivalence structures, Galois functoriality, and comparison intervals*, preprint (2026), archived at DOI 10.5281/zenodo.22823772.

[17] R. Shibusawa, *Joins of localisations in the lattice of two-out-of-three classes: the finite-iteration criterion for closure operators on exact categories, with applications to toposes, torsion theories and fuzzy sets*, preprint (2026), archived at DOI 10.5281/zenodo.23691896.

[18] R. Shibusawa, *Comparing theories in the lattice of two-out-of-three classes: Morita equivalence as a join of completions, the forced order of coproducts and quotients, and conceptual completeness as an identity of samenesses*, preprint (2026), archived at DOI 10.5281/zenodo.24095158.

[19] D. Tsementzis, *A syntactic characterization of Morita equivalence*, J. Symbolic Logic 82 (2017), 1181–1198; statements are numbered as in the preprint arXiv:1507.02302v1.

Daiichi Institute of Technology
Email address: r-shibusawa@daiichi-koudai.ac.jp